Appendix

A Message Passing for Response Embeddings

Given user and response embeddings at layer ℓ, a message from neighborhood users to the response as

$$\boldsymbol{m}_r^+ = \sum_{u \in \mathcal{N}_r^+} \alpha_{u,r} \Big(\hat{W}_1^{(\ell)} \boldsymbol{e}_u^{(\ell)} + \hat{W}_2^{(\ell)} (\boldsymbol{e}_u^{(\ell)} \odot \boldsymbol{e}_r^{(\ell)}) \Big),$$
$$\boldsymbol{m}_r^- = \sum_{u \in \mathcal{N}_r^-} \beta_{u,r} \Big(\hat{W}_3^{(\ell)} \boldsymbol{e}_u^{(\ell)} + \hat{W}_4^{(\ell)} (\boldsymbol{e}_u^{(\ell)} \odot \boldsymbol{e}_r^{(\ell)}) \Big),$$
$$\boldsymbol{m}_r^{(\ell)} = \hat{W}_{\text{self}}^{(\ell)} \boldsymbol{e}_r^{(\ell)} + \boldsymbol{m}_r^+ + \boldsymbol{m}_r^-, \tag{12}$$

where $\hat{W}_1^{(\ell)}, \hat{W}_2^{(\ell)}, \hat{W}_3^{(\ell)}, \hat{W}_4^{(\ell)}, \hat{W}_{\text{self}}^{(\ell)} \in \mathbb{R}^{d \times d}$ are parameter matrices, $\odot$ is element-wise multiplication, and $\alpha_{u,r}$ and $\beta_{u,r}$ are normalization factors, set to $\frac{1}{\sqrt{|\mathcal{N}_u^+| \cdot |\mathcal{N}_r^+|}}$ and $\frac{1}{\sqrt{|\mathcal{N}_u^-| \cdot |\mathcal{N}_r^-|}}$, respectively.

Then, the response embedding is updated with the aggregated message $\boldsymbol{m}_r^{(\ell)}$:

$$\boldsymbol{e}_r^{(\ell+1)} = \psi\big(\boldsymbol{m}_r^{(\ell)}\big), \tag{13}$$

where $\psi(\cdot)$ is a non-linear activation.

B Method Baselines

Uniform. The uniform model is a standard approach for pairwise preference comparisons. We train the uniform model with all annotation pairs, which will capture the common preference.

Oracle. For an oracle model of our setting, we train the model with the true group membership of all users. A separate uniform model is trained for each group by aggregating annotations from the users in that group.

I2E (Li et al., 2024). I2E is a framework that uses DPO to personalize LLM. However, it can be easily extended to reward modeling. I2E trains a model that maps the user index into a learnable embedding. It appends each user embedding as an additional input token to the LLM, providing user-specific signals for reward prediction.

I2E$_{\text{proxy}}$ (Li et al., 2024). A variant of I2E that introduces N proxy embeddings. A weighted combination of these proxies forms the final user embedding, which is passed to the LLM for reward prediction. In our experiments, we use $N = 10$.

VPL (Poddar et al., 2024). Variational Preference Learning (VPL) encodes user-specific annotations into user embeddings. The user embeddings are then combined with sentence representations via an MLP to predict reward scores. To capture the user preferences effectively, VPL uses a variational approach that maps the user annotations into a prior distribution.

PAL (Chen et al., 2024a). Pluralistic Alignment (PAL) applies an ideal-point model, where the distance between the user and the response determines the reward. The ideal point of the user is represented by N proxies, set to $N = 10$ in this work. Among variants of PAL, we use PAL-A with logistic loss.

C Related Works

Personalized alignment. With the growth of generative models, alignment has emerged as a crucial strategy for mitigating undesirable outcomes, such as biased or harmful outputs, and ensuring that the model works with human preference (Dai et al., 2023; Yang et al., 2024a). Alignment methods often rely on reward models. They typically build on the BTL framework, which relies on pairwise comparisons from various annotators. However, previous research has often focused on the average preference of annotators (Achiam et al., 2023), ignoring the diverse preferences.

To address preference diversity, recent works (Jang et al., 2023; Oh et al., 2024; Yang et al., 2024b) view this problem as a soft clustering problem, where user-specific preferences are treated as mixtures of predefined preference types. Although this approach effectively handles diverse preferences, it relies on specifying several preference types in advance.

Another line of work introduces user latent variable in the BTL framework (Poddar et al., 2024; Li et al., 2024; Chen et al., 2024a). Although extending the BTL framework with latent user variables can address diverse preferences, the main challenge lies in obtaining user representations. One approach is to treat each user embedding as learnable parameters, (Li et al., 2024; Chen et al., 2024a), and the other strategy is to train an encoder that infers embeddings from the small set of annotated pairs provided by each user (Poddar et al., 2024).

Preference learning with sparse interactions. Preference learning with sparse interactions is a well-studied challenge in recommendation systems, where each user typically interacts with only a small fraction of the available items. Despite these limited interactions, the system should infer the preference of each user and recommend additional items accordingly (He and Chua, 2017; Chen et al., 2020; Li et al., 2022; Lin et al., 2022). Collaborative filtering (CF) is a widely adopted solution that assumes users with similar interaction histories will exhibit similar preferences.

Graph-based CF (GCF) (Wang et al., 2019; He et al., 2020) has been considered one of the most advanced algorithms for a recommendation system. GCF leverages graph neural networks (GNNs) to capture preference through the connectivity among users and items. Many GCFs are developed based on an implicit feedback assumption (Rendle et al., 2012), where an edge between a user and an item reveals a preferable relation. Whereas in our setting, users provide explicit feedback given a pair of responses, making direct application of GCF unsuitable.

D Experimental Details

In this section, we provide a detailed explanation of dataset construction and hyper-parameters.

D.1 Datasets

TL;DR. The TL;DR dataset (Stiennon et al., 2020) contains Reddit posts alongside concise summaries and annotator IDs. Prior works (Li et al., 2022; Chen et al., 2024a) employ a modified version of this dataset by defining two simulated preference groups: one group favors shorter summaries, while the other prefers longer ones. The two groups provide different annotations for each summary pair. To focus on the most active annotators, they retain only the ten users with the highest number of annotations. We adopt the resulting set of annotation pairs from these ten users as our survey set.

Ultrafeedback-P. Poddar et al. (2024) proposes the Ultrafeedback-P (UF-P) benchmark for personalized reward modeling, based on the Ultrafeedback (UF) dataset (Cui et al., 2023), which provides response pairs rated on four attributes: helpfulness, honesty, instruction following, and truthfulness. In UF-P, each attribute corresponds to a distinct preference. For instance, a user belonging to the helpfulness group annotates pairs, solely considering the helpfulness score.

UF-P-2 employs only two attributes and removes pairs that both user groups label identically, focusing on controversial cases where preferences differ. In **UF-P-4**, all four attributes are retained as preference dimensions, which allows for partial agreement among groups and hence increases complexity. Although Poddar et al. (2024) also excludes pairs fully agreed upon by all users, the remaining set is larger and exhibits more variety than UF-P-2.

In Poddar et al. (2024), each user is given a small context sample from a limited set of unannotated pairs to infer the user's preference. In contrast, we leverage every available pair in the dataset to infer each user's preferences. For our dataset construction, we use UF-P-4 dataset.

PersonalLLM. PersonalLLM (Zollo et al., 2024) is built with 10,402 open-ended prompts that were sampled from a larger pool of 37,919 conversational questions drawn from public RLHF and preference benchmarks such as Anthropic HH-RLHF (Bai et al., 2022), NVIDIA HelpSteer (Wang et al., 2023), and RewardBench (Lambert et al., 2024). For each prompt, they used eight frontier chat models to generate a diverse response set that minimizes obvious quality gaps while covering latent preference dimensions. The resulting (prompt, response1, response2, ..., response8) tuples are split into 9,402 training and 1,000 test items.

Each response is evaluated by ten strong open-source reward models with heterogeneous alignment objectives. These reward models assign scalar scores capturing distinct value dimensions for every response. Storing the full 10×8 matrix of scores per prompt provides a dense, model-agnostic preference signal that later steps can recombine to reflect arbitrary preferences. To simulate a large user base, they treat the preference of a user as a weighted ensemble over the ten reward models. The weight is sampled from a Dirichlet distribution, where varying the concentration parameter controls preference diversity.

We use $\alpha = 0.1$ for Dirichlet distribution. Due to computational constraints, we simplify the dataset by selecting three responses per prompt and considering only four reward dimensions. Following Poddar et al. (2024), we remove *non-controversial* response pairs—those in which one response is strictly ranked below the other across

all preference dimensions—to ensure the heterogeneity.

D.2 Hyper-parameters

We describe the training details of GNN, a reward model, and unseen user adaptation, such as model architecture and hyper-parameters.

GNN. The model consists of four message-passing layers, each with user and response embeddings of dimension 512. We use Leaky ReLU as a non-linear activation function to update user and response embeddings. Training proceeds for 300 epochs using the AdamW optimizer (Loshchilov, 2017) with a learning rate of 1×10^{-4} and a cosine scheduler with warmup ratio 0.1. The batch size is 1024, and all experiments are conducted on an RTX 4090 GPU.

Reward models. CoPL comprises an LLM backbone and a MoLE adapter. We use `gemma-2b-it` or `gemma-7b-it` as the LLM backbone. MoLE includes one shared expert and eight LoRA experts with a rank of eight. A two-layer MLP with a hidden dimension of 256 and ReLU activation serves as the gating mechanism, with a temperature set to 1.

We train the reward models using the AdamW optimizer with a learning rate of 5×10^{-5} and a cosine scheduler with warmup ratio 0.03. Four GPUs, such as RTX6000ADA, L40S, and A100-PCIE-40GB, are employed with a batch size of 32 per GPU for `gemma-2b-it` and 16 per GPU for `gemma-7b-it`.

Baseline models use LoRA with rank 64. They also trained with an AdamW optimizer and a cosine scheduler with a warmup ratio 0.03. We search the learning rate from $[1\times10^{-4}, 5\times10^{-5}, 1\times10^{-5}, 5\times10^{-6}]$.

User adaptation. We use a two-hop seen user and 0.07 as temperature for unseen user adaptation of CoPL. For I2E, each learnable user representation is mapped into each user. For $\text{I2E}_{\text{proxy}}$ and PAL, user representations are determined by $N = 10$ proxies. Adapting to an unseen user requires parameter optimization for unseen users, typically through several gradient steps. To optimize the parameters for unseen users, 50 gradient steps are applied during adaptation.

E Additional Experimental Results

Performance under the imbalanced group distribution with UF-P-4 (AVG). Table A2 reports group-wise accuracies for the four group UF-P-4 (AVG) setting under selected imbalance configurations. The results exhibit the same trend seen in the two-group setting. CoPL continues to capture diverse user preferences across all groups. As the distribution departs from the balanced 1:1:1:1 setting, the gap from the balanced baseline widens. The lower absolute accuracy of some groups is largely due to the intrinsic difficulty of their preferences rather than the imbalance itself. This interpretation is supported by the G-Oracle. Fig. A6 visualizes the learned user embeddings. The embeddings form well-separated clusters aligned with group identities even under strong imbalance, which suggests that the representation remains stable, although predictive performance on minority groups drops.

Ablation study of the number of users. Fig. A3 shows that CoPL performs robustly across different expert counts. This indicates that a moderate number of experts is generally sufficient to capture diverse user preferences.

Performance with large-scale LLM. To assess scalability, we instantiate CoPL with gemma-2-27B-it (Team et al., 2024b) and evaluate on UF-P-4 (ALL) and PersonalLLM (ALL). We use a single seed due to hardware limits and compare with VPL, the strongest baseline. As shown in Table A3, CoPL surpasses VPL on both datasets, indicating that the gains carry over to larger model scales. These results support the scalability of CoPL beyond the settings used in the main experiments.

Ablation study of message-passing. Inspired by the previous work (He et al., 2020) in recommendation systems, we first omit the non-linear activation and feature transformation matrix used in Eq. (2), and also investigate the effectiveness of negative edges. As shown in Table A4, incorporating negative edges consistently improves accuracy. Notably, our proposed message-passing achieves the highest accuracy, highlighting both the effectiveness of our message-passing operation and the advantage of modeling negative edges.